

# Same synergy. Different added information.

Finite categorical MGW shared exclusions · signed population atoms · nats

**Shared law: A, B and U are mutually independent fair bits.**

Baseline A; target  $Y = (A, B)$ ; added candidate C is U or B.

Both source pairs have  $P(A = a, C = c) = 1/4$  for  $a, c$  in  $\{0, 1\}$ .

| Signed MGW atom                       | Add C = U<br>Unrelated to target Y | Add C = B<br>Missing target bit |
|---------------------------------------|------------------------------------|---------------------------------|
| Redundancy                            | $\ln(4/3)$                         | $\ln(4/3)$                      |
| Unique baseline A                     | $\ln(3/2)$                         | $\ln(3/2)$                      |
| Unique added C                        | $-\ln(4/3)$                        | $\ln(3/2)$                      |
| Net synergy S                         | $\ln(4/3)$                         | $\ln(4/3)$                      |
| Added information<br>$I(C; Y \mid A)$ | 0                                  | $\ln 2$                         |

**Unique added C + net synergy = added conditional MI**

U:  $-\ln(4/3) + \ln(4/3) = 0$

B:  $\ln(3/2) + \ln(4/3) = \ln 2$

Synergy ties; the full signed decomposition retains the CMI difference.

Signed cancellation is published in MGW (RndErr, §VI.3 / Table III).

Matched comparison: priority unestablished; no estimator or deployment claim.